

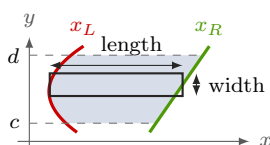
L4.1c Areas Along y

Sideways rectangles — what you need to know

Upright or sideways

	Upright rectangle	Sideways rectangle
length	$y_T - y_B$	$x_R - x_L$
width	Δx	Δy
boundaries written as	y in terms of x	x in terms of y
limits are	x -coordinates of the crossings	y -coordinates of the crossings
which curve is which	test an x : the larger y is on top	test a y : the larger x is on the right
area	$\int_a^b y_T - y_B dx$	$\int_c^d x_R - x_L dy$

AREA WITH SIDEWAYS RECTANGLES



$$A = \int_c^d x_R - x_L dy$$

both boundaries continuous on $[c, d]$, written as x in terms of y

$$\text{length} = x_R - x_L, \quad \text{width} = \Delta y, \quad A = \lim_{n \rightarrow \infty} \sum_{i=1}^n (x_R - x_L) \Delta y$$

“each rectangle is as long as the gap between the curves and as tall as Δy ”

- ① **Solve each boundary for x .**
- ② **Get the limits.** Solve $x_R = x_L$ — they are the y -coordinates of the crossings.
- ③ **Test a value of y** to decide which boundary is on the right. Evaluate both there and compare.
- ④ **Integrate right minus left.** If the curves trade places, one integral per piece.

WHICH RECTANGLE

Draw one. If both its ends stay on one curve each all the way across the region, that is your variable. Upright: $y_T - y_B dx$. Sideways: $x_R - x_L dy$.

Two reasons to split, and they are different.

A boundary turns around in the interval x_L and x_R **cross** in the interval
a *graphing* problem — split to *write* it as $x =$ an *area* problem — split the *integral*: right
 $g(y)$, one branch per piece. The area may still and left trade places, exactly as top and bot-
be one integral tom did

“split at the crossing, not at the turn”

WATCH OUT

~~✗ the limits are the x -coordinates of the crossings~~ → the y -coordinates — you are integrating dy

~~✗ x_R is the bigger-looking expression~~ → test a y inside the interval; the larger x is on the right

~~✗ a negative area~~ → you subtracted left minus right — it is $x_R - x_L$

~~✗ not monotonic, so it cannot be integrated in y~~ → the integral only needs every horizontal line to cross the region once

~~✗ split at the turning points~~ → split at the crossings, where x_L and x_R trade places